Department of Computer Science
& Information Technology

Course Information	
Course Code: **CS-08519**	Course Title: **Computer Networks**
Credit Hours: 4 **(3,1)**	Lecture Hours/Week: **2**
Lab Hours/Week: **3**	Pre-Requisites: **None**

a. Course Objectives:

1. To develop a comprehensive ***understanding*** of computer network fundamentals, including architectures, components, topologies, and communication models such as OSI and TCP/IP, to establish a solid foundation for modern data communication systems.

2. To ***explain*** physical layer concepts and transmission techniques, focusing on analog and digital signals, line coding, noise, attenuation, and capacity formulas, while understanding multiplexing, switching, and Ethernet technologies in both wired and wireless environments.

3. To ***analyze*** the functions of data link and network layers, including framing, error detection and correction, flow control, addressing, subnetting, and routing protocols such as RIP, OSPF, and EIGRP for efficient packet delivery.

4. To comprehend the operation of transport and application layer protocols, including TCP, UDP, SCTP, HTTP, FTP, DNS, DHCP, SMTP, IMAP, and POP3, and their roles in ensuring reliable end-to-end communication and Internet services.

5. To ***identify*** and address network security challenges, through understanding encryption, authentication, and secure protocol design, ensuring confidentiality, integrity, and availability of information within communication networks.

6. To gain practical experience in network ***design and troubleshooting***, by configuring LANs and WANs using Cisco Packet Tracer, implementing routing and switching, and evaluating network performance through laboratory experiments and simulations.

7. To explore emerging trends in networking technologies, such as IoT, 5G, cloud-based networking, virtualization, and Software Defined Networking (SDN), to understand their impact on next-generation communication systems.

8. To prepare students for professional and research-oriented roles in the field of computer networks and cybersecurity by integrating theoretical knowledge with analytical, design, and implementation skills.